

Welcome to **instats**

The session will begin shortly.

Start

A 3-DAY WORKSHOP ON LLMS FOR MEASUREMENT

Generative LLMs for wrangling and crawling.

Free-text extraction, polite crawling, and an end-to-end pipeline. Same reproducibility demands as Day 2. New tool.

INSTRUCTOR

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PREREQUISITE

Day 2 hands-on cell finished running

DURATION

2 hours 30 minutes

Four movements, from one prompt to a pipeline.

MOVEMENT 01 • 17 MIN

Why generative

The decision rule. When the encoder runs out of road and the generative tool becomes the right one.

MOVEMENT 02 • 35 MIN

Structured extraction

Fifty news leads, four JSON fields, prompt caching, and validation that does not need any hand-coding.

MOVEMENT 03 • 35 MIN

Lightweight crawling

Robots.txt live on screen. A polite Python fetcher. Raw HTML stored for reproducibility.

MOVEMENT 04 • 25 MIN

End-to-end pipeline

Crawl, parse, extract, validate, save. Run together on five fresh articles. One full row of output.

One ten-minute break after Movement 2. Closing reproducibility checklist in the last fifteen minutes.

Days 1 and 2 in one slide.

DAY 01

The model zoo

Two families of LLMs. Encoders read text and return labels. Generative models continue text. Day 1 ran zero-shot DeBERTa on five paragraphs of US trade-policy text.

DAY 02

From a model to a measurement

Codebook, thirty-paragraph gold set, thirteen-hypothesis ensemble, country-year proxy plotted against external events, and a validation appendix.

DAY 03

The other family

Today we leave the encoder behind. We pick up the generative model and use it for tasks the encoder cannot easily do.

The encoder is the right tool for closed labels. There are tasks for which it is not. Today is about those tasks.

Three jobs the encoder cannot easily do.

JOB 01

Free-text extraction

"France and Kenya signed a bilateral health-sector aid agreement worth EUR 80 million."

Encoder can tell you it mentions a signing. It cannot tell you who signed, or for how much.

JOB 02

Normalization

"USA," "United States," "U.S.," "the United States of America."

Encoder labels each. Generative resolves them all to a single canonical form in one prompt.

JOB 03

Light summarization

A 600-word press release with three buried facts about the agreement.

Encoder needs a fixed schema of labels in advance. Generative picks out the three facts on first read.

Same reproducibility demands as yesterday. New tool. The intuition carries over.

FIRST MOVEMENT • 17 MINUTES

Why reach for generative.

The decision rule that tells you when the encoder runs out of road.

01

One question tells you which family fits.

DIAGNOSTIC 01

Closed list?

Can the answer be expressed as a value from a finite list you wrote down before reading any text?

Yes. Encoder.

No. Generative.

DIAGNOSTIC 02

Messy input?

Is your input messy in a way an encoder hypothesis cannot fix? Country-name variants, date formats, alias resolution.

Yes. Generative.

DIAGNOSTIC 03

Variable schema?

Do you need three or four pieces of information that may or may not appear, with different surface forms each time?

Yes. Generative.

If two of three are yes, the generative tool is almost certainly the right starting point. If two of three are no, run an encoder first and reach for a generative model only if the encoder ceiling is hit.

Tradeoffs that go in your replication appendix.

COST 01

Per-token billing

Generative API calls are billed per input and output token. Today's demo on Anthropic Haiku 4.5 costs roughly five US cents for fifty leads plus five end-to-end articles.

Fine for a paper. Becomes a research-design constraint at million-paragraph corpus scale.

COST 02

Non-determinism

Even at temperature zero, two calls with the same prompt can produce slightly different outputs. Providers also retire models behind the scenes.

Pin model version. Run the determinism cell. Report the number.

COST 03

Prompt versioning

The prompt is a new artifact in your replication package. If the prompt changes, the output changes. Same for the model version.

Ship system prompt verbatim. Include date finalized. Hash it for the audit log.

None of the three costs existed in yesterday's encoder appendix. All three are now part of the methods burden.

When to reach for which family.

	CLOSED LABELS	FREE-FORM, MESSY, OR VARIABLE
Large corpus tens of thousands+	ENCODER, FINE-TUNED Yesterday's pipeline. Hand-coded gold, weighted ensemble. Determinism by construction.	GENERATIVE + ENCODER HYBRID Generative on upstream extraction. Encoder on downstream classification. Both validation tracks.
Small corpus hundreds to a few thousand	ENCODER, ZERO-SHOT Day 1's pipeline. NLI hypotheses. Cheap, deterministic. Often enough on its own.	GENERATIVE, PROMPT-ONLY Today's pipeline. Schema, system prompt, validators, audit log. Iteration speed for the win.

Most political-science measurement projects sit in the bottom-left cell. Today is about the right column. Hybrid pipelines combine them.

SECOND MOVEMENT • 35 MINUTES

Structured extraction.

Fifty news leads. Four JSON fields. The single most common pattern political scientists need from a generative model.

02

Decide the schema before you touch any code.

```
{ "signatories": list[str], # parties to the agreement  
  "agreement_type": enum, # closed taxonomy, 12 values  
  "year": int | null, # 1990-2030 or null "sector":  
  enum, # closed taxonomy, 13 values }
```

THREE PROPERTIES

- **Finite.** The field set is decided in advance. Adding a new field after the fact means re-running every call.
- **Typed.** Every field has a type and, where possible, a closed taxonomy. The validator in section 5 of the notebook reads off the type signatures.
- **Pinned.** The schema is documented in the system prompt and in the replication package. Future-you needs both.

System prompt holds the schema. User prompt holds the lead.

SYSTEM PROMPT • CACHED • SAME EVERY CALL

You are an information extraction system. Extract a JSON object with exactly four fields.

"signatories": list of strings, normalized to short country names.

"agreement_type": one of [bilateral_trade, multilateral_trade, climate, defense, health, financial_assistance, IP, security_cooperation, infrastructure, technology, other].

"year": four-digit integer or null.

"sector": one of [trade, climate, defense, health, finance, IP, security, infrastructure, technology, energy, agriculture, other].

Return ONLY the JSON. No markdown.

Example output:

```
{"signatories": ["France", "Kenya"], "agreement_type": "health", "year": 2024, "sector": "health"}
```

Cached at ~10% of base rate across 50 calls. The example is the highest-leverage line.

USER PROMPT • PER ROW • NOT CACHED

"On 8 September 2020, the United Arab Emirates and Israel signed the Abraham Accords, normalizing diplomatic relations and establishing cooperation in security, technology, and tourism."

One news lead per call. Variable. Not cacheable.

Switch to `demo.ipynb` sections 1 to 4.

Run together

Section 1. Mount Drive. Install Anthropic SDK. Pull `ANTHROPIC_API_KEY` from `userdata`.

Section 2. Load `news_leads.csv`. Inspect three sample leads.

Section 3. Define the schema. Print the system prompt and the closed taxonomies.

Section 4. Run extraction on a single lead. Inspect the parsed JSON, input and output token counts, and any cache-related counters in the response.

If the API call returns a 401, double-check that the key is saved under the key icon in Colab and that the cell pulled it via `userdata.get()`.

Four checks you get for free without writing any gold.

CHECK 01

Schema validation

Every output parses as JSON. Every required field is present. Every value matches its declared type or taxonomy. Programmatic.

CHECK 02

Determinism

Same prompt, two runs, temperature zero. Count identical outputs. The drift you see is the floor on reproducibility.

CHECK 03

Sample audit

Read five random outputs by hand. Ten minutes of human time. Smallest defensible spot-check.

CHECK 04

Audit log

One row per call. Lead, prompt hash, timestamp, response, token counts, cost. The replication artifact.

Necessary, but not sufficient. These four checks are prerequisites for any further validation. They do not replace a hand-coded gold set. Slide 28 is where the gold set comes back as the missing piece for a publishable paper.

What you should expect to see when the cells run.

Schema pass rate

~98%

50 leads, four-field schema with one example. Modern Haiku-class models clear the high-90s on first pass.

Determinism

~9 / 10

10-lead subset, two runs, temperature 0. The drift comes from leads with ambiguous sector tagging.

Audit rows

50

One per call. System-prompt hash identical across all 50 because the prompt is cached.

Estimated cost

~\$0.01

50 calls at Haiku 4.5 pricing with prompt caching. Well within the few-cents per replication budget.

Numbers are illustrative based on typical runs. The exact values from the live demo will appear in the notebook output. If Anthropic ships a model update between sessions, expect the determinism number to shift before you re-pin.

From a schema to an audit log in five notebook sections.

WHAT YOU BUILT

- A four-field schema with closed taxonomies pinned in the system prompt.
- A two-part prompt with a stable system message and a per-row user message.
- Fifty extraction calls with prompt caching at ~10% of base input rate.
- Programmatic schema validators and a determinism check.
- A five-record sample audit and a full audit log with cost.

WHAT IS STILL MISSING

- A hand-coded gold set with field-level F1 against truth.
- A subgroup breakdown of pass rate. Sector, year, signatory count.
- A simpler-baseline comparison. Rule-based extraction or a smaller model.
- A cost projection at production scale.

Slide 28 brings these back. Today's pipeline is the template you reuse when you have the budget for one.

Stop